Industrial Internship Report on

”Music Player using JAVA”

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time.

My project was The **Java Music Player** is a desktop application developed using Java and Java GUI. It allows users to play audio files through a simple and user-friendly interface. The application includes features such as play, pause, and stop for controlling music playback. Users can create playlists within the application and also load playlists from external files. This project demonstrates the use of Java GUI components, event handling, and file management in building a multimedia application.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

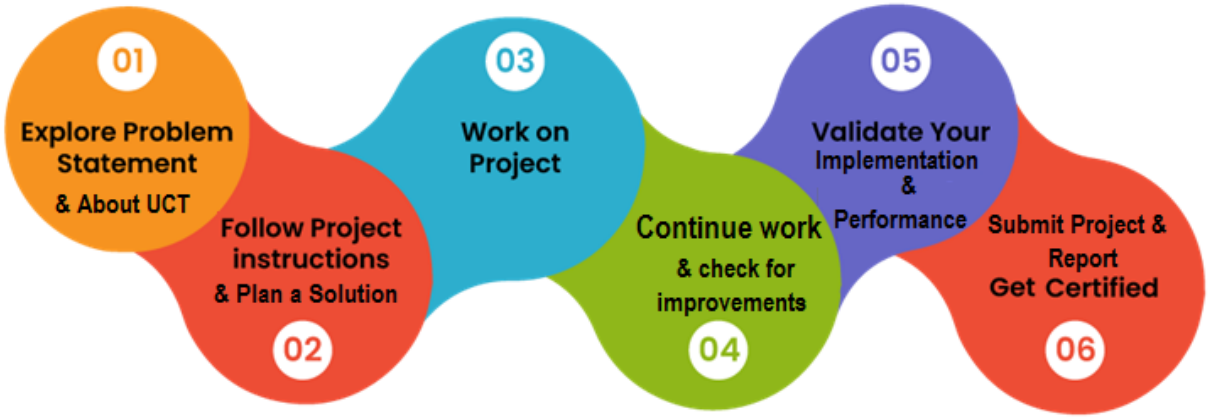
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

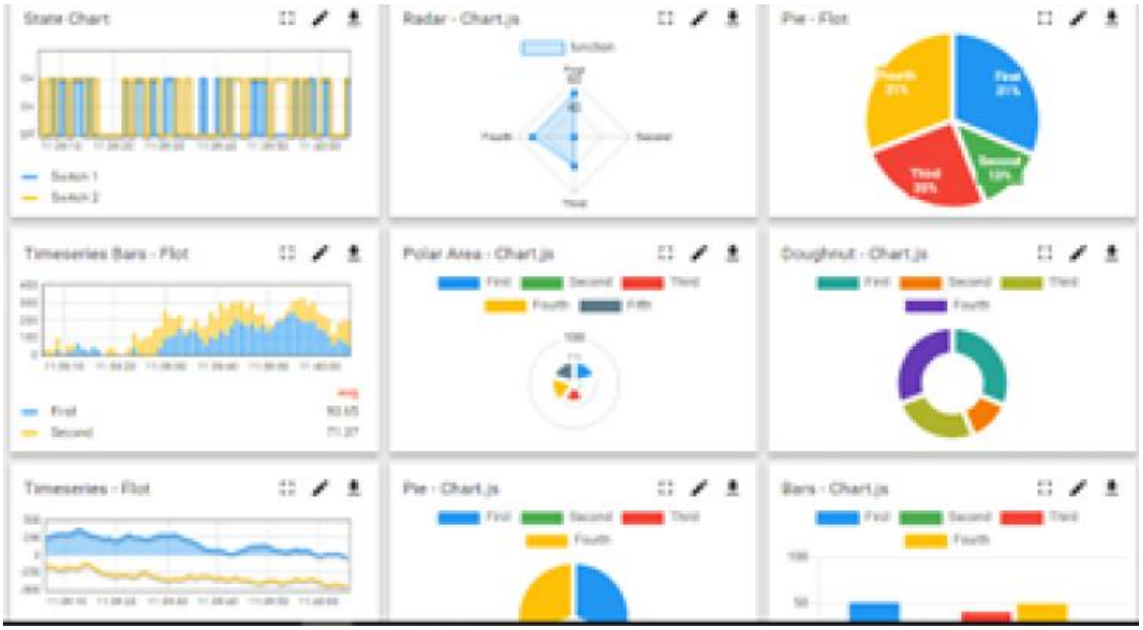
i. UCT IoT Platform (uct Insight)

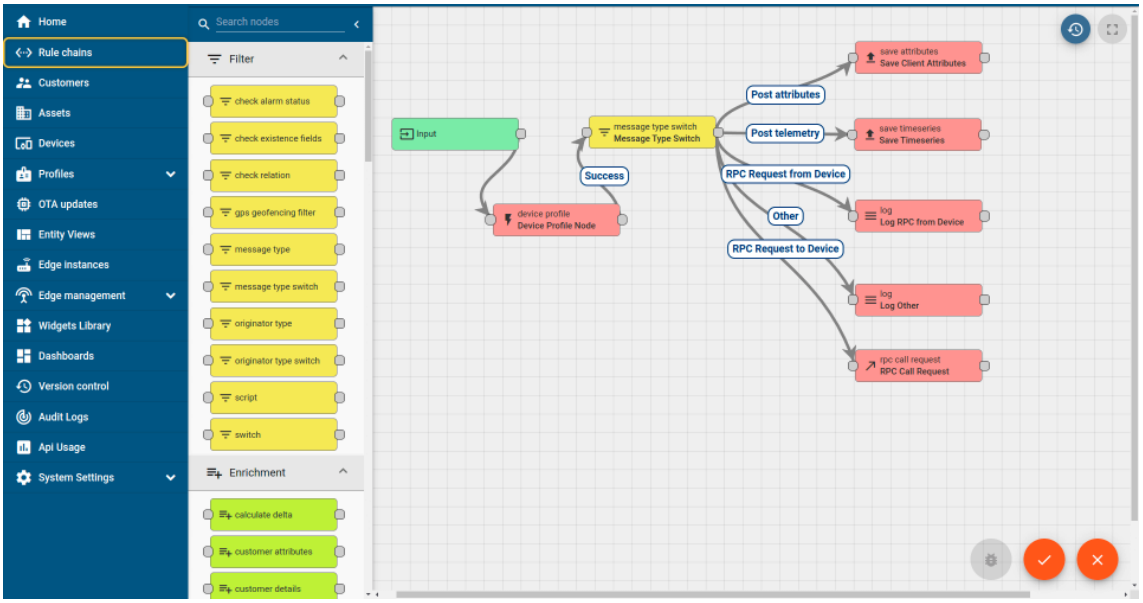
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

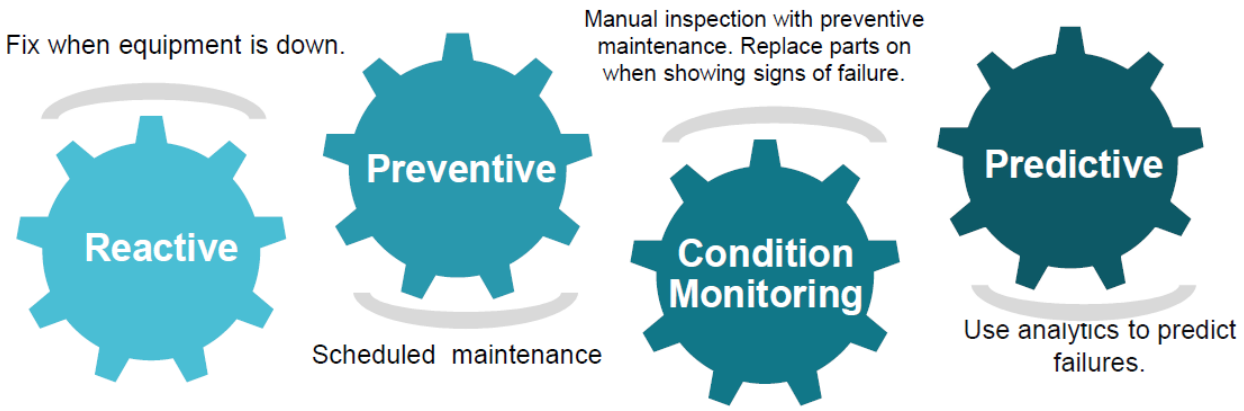
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

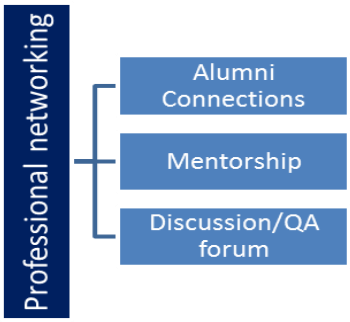
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

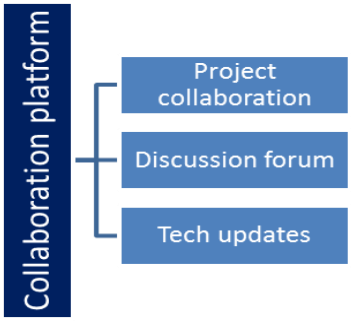


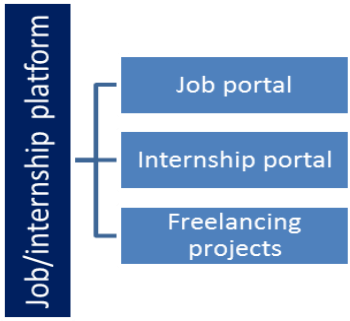
2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

3 Problem Statement

In the assigned problem statement, the objective is to develop a **simple music player application using Java** that allows users to easily play and manage audio files. Many basic media players do not provide a simple interface for creating and managing playlists. The system should allow users to **play music, create playlists within the application, and load playlists from external files**. The goal of this project is to build a **user-friendly Java GUI application** that helps users organize and enjoy their music efficiently. In the assigned problem statement, the objective is to develop a **simple music player application using Java** that allows users to easily play and manage audio files. Many basic media players do not provide a simple interface for creating and managing playlists. The system should allow users to **play music, create playlists within the application, and load playlists from external files**. The goal of this project is to build a **user-friendly Java GUI application** that helps users organize and enjoy their music efficiently.

4 Existing and Proposed solution

Existing Solutions:
Many existing music players allow users to play audio files and manage playlists. However, some of these applications are complex, require installation of large software packages, or have limited customization options for playlist management. In addition, some basic players do not allow users to easily create and load playlists from external files.

Limitations:

- Complex user interfaces that are not beginner-friendly
- Limited playlist management features
- High system resource usage in some applications
- Lack of simple customization for users

Proposed Solution:
The proposed solution is to develop a **Java-based Music Player** with a simple and user-friendly graphical interface. The application allows users to play audio files, create playlists directly in the application, and load playlists from external files. The system focuses on simplicity and efficient music management.

Value Addition:

- Simple and clean Java GUI interface
- Ability to create playlists within the application
- Option to load playlists from external files
- Lightweight application with easy navigation

4.1 Code submission : https://github.com/ashwinrlab/upskillcampus.git

4.2 Report submission : https://github.com/ashwinrlab/upskillcampus.git

5 Proposed Design/ Model

The proposed system is a **Java-based Music Player application** designed to provide an easy way for users to play and manage music files. The system follows a simple design flow that starts with user interaction through a graphical interface and ends with audio playback.

First, the user opens the application and interacts with the **Graphical User Interface (GUI)**. Through the interface, the user can select music files, create playlists, or load existing playlists from external files. The system processes the user input and sends it to the **music control module**, which handles functions such as play, pause, and stop.

In the intermediate stage, the application manages the selected audio files and organizes them into playlists. The playlist manager maintains the list of songs and allows users to navigate between tracks.

Finally, the **audio playback module** reads the selected audio file and plays it through the system's audio output. The user can control playback using the interface buttons.

- **5.1 High Level Diagram (if applicable)**

The high-level diagram represents the overall working structure of the system. It shows how the **User Interface**, **Playlist Manager**, and **Audio Playback Module** interact with each other to provide music playing functionality.

Figure 1: High Level Diagram of the System

User → Graphical User Interface → Playlist Manager → Audio Playback Module → Audio Output

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- **5.2 Low Level Diagram (if applicable)**

The low-level diagram explains the internal components of the application in more detail. It includes modules such as:

- **User Interface Module** – Handles user interaction through buttons and menus
- **File Loader Module** – Loads music files and playlist files from the system
- **Playlist Manager** – Stores and manages the list of songs
- **Audio Control Module** – Controls play, pause, and stop functions

- **Audio Playback Module** – Processes and plays the selected music file

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- **5.3 Interfaces (if applicable)**

The system includes a **Graphical User Interface (GUI)** that allows users to interact with the application easily. The interface contains buttons and controls such as:

- Play Button
- Pause Button
- Stop Button
- Create Playlist Option
- Load Playlist Option
- Song List Display

The data flow begins when the user selects a song or playlist through the interface. The system then processes the request and sends it to the audio module, which plays the selected file. The application uses **Java event handling and file management mechanisms** to manage user input and audio playback efficiently.

6 Performance Test

Performance testing is conducted to evaluate how efficiently the **Java Music Player application** performs under different conditions. The goal is to ensure that the system runs smoothly while playing music files and managing playlists without consuming excessive system resources.

One of the main constraints considered in this project is **memory usage**, since multimedia applications can consume significant memory when handling multiple audio files. Another constraint is **application speed and responsiveness**, ensuring that user actions such as play, pause, and playlist loading respond quickly. The design of the application focuses on lightweight processing and efficient file handling to reduce memory usage and improve performance.

The system was tested to ensure that music playback occurs without delay and that playlists can be loaded and managed smoothly. The application performs well for normal usage with standard audio files. Even though extensive industrial performance testing tools were not used, the design considers possible limitations and suggests improvements such as optimized file handling and better memory management for larger music libraries.

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7 6.1 Test Plan / Test Cases

Test Case	Description	Expected Result
TC1	Launch the application	Application opens successfully
TC2	Load a music file	Selected file loads without errors
TC3	Play music	Audio plays correctly
TC4	Pause music	Audio playback pauses
TC5	Stop music	Music playback stops completely
TC6	Create playlist	Playlist is created and songs are added
TC7	Load playlist from file	Playlist loads successfully

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8 6.2 Test Procedure

1. Install and run the Java Music Player application.
2. Select an audio file from the system storage.
3. Click the **Play** button to start playback.
4. Use **Pause** and **Stop** buttons to control playback.
5. Create a new playlist and add multiple songs.
6. Load a playlist from an external file and verify that all songs are displayed.
7. Observe the system performance during playback and playlist operations.

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9 6.3 Performance Outcome

The testing results show that the **Java Music Player** performs effectively for playing audio files and managing playlists. The application responds quickly to user inputs such as play, pause, and stop. Memory usage remains low since the system loads only the required audio file during playback. Overall, the application demonstrates stable performance and meets the requirements of a simple multimedia player.

10 My learnings

During the development of the **Java Music Player project**, I gained practical experience in building a desktop application using the Java programming language. I learned how to design and implement a **Graphical User Interface (GUI)** and handle user interactions using event handling techniques. This project also helped me understand **file handling and playlist management** for multimedia applications. Additionally, I improved my problem-solving skills while implementing features such as music playback and playlist loading. Overall, this project enhanced my understanding of software development and will help me build more advanced applications in the future.